Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science student at Sapienza Università di Roma specializing in Machine Learning and Artificial Intelligence. Experienced in developing machine learning solutions, evaluating model performance, automating data workflows, and deploying AI applications using Python, SQL, FastAPI, Docker, and AWS. Strong foundation in statistical analysis, classification, regression, NLP, computer vision, and data-driven problem solving. Published IEEE researcher with hands-on experience translating analytical concepts into practical machine learning solutions. Seeking Data Scientist, Machine Learning Engineer, and AI-focused roles.

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Deep Learning and Applied AI, NLP, Computer Vision, Distributed Systems.
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

TECHNICAL SKILLS

- **Programming :** Python, Bash, SQL, C++
- **Data Science & Machine Learning :** Machine Learning, Statistical Analysis, Classification, Regression, Feature Engineering, Model Evaluation, Cross Validation, Data Cleaning, Exploratory Data Analysis
- **Libraries & Frameworks :** Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, Keras, Sentence Transformers
- **Deployment & Cloud :** FastAPI, REST APIs, Docker, AWS ECS, AWS Fargate, CloudWatch
- **Tools :** Git, GitHub, Linux

PROJECTS

DocDelta [Document Comparison API]

- Developed an end-to-end NLP solution for semantic comparison of legal and contract documents using Sentence Transformers and cosine similarity.
- Designed algorithms to identify added, removed, and modified content while distinguishing numerical changes from textual modifications.
- Built an automated document comparison pipeline that reduced manual review effort and enabled scalable processing of document revisions.
- Deployed the application using FastAPI, Docker, AWS ECS Fargate, and CloudWatch for scalable inference and monitoring.
- Evaluated inference performance and optimized deployment architecture, achieving approximately 840ms warm inference latency.

Lossy Text Compressor [Semantic NLP Pipeline]

- Developed a machine learning pipeline using GTR-T5 embeddings, vector quantization, and text reconstruction to analyze compression-quality trade-offs.
- Performed systematic experiments across multiple quantization levels and evaluated results using cosine similarity and ROUGE-L metrics.
- Achieved optimal performance at 6-bit quantization with cosine similarity of 0.964 and ROUGE-L score of 0.776.
- Identified non-linear relationships between compression levels and reconstruction quality, providing insights for efficient semantic storage and retrieval.
- Conducted quantitative analysis and model evaluation to determine optimal compression-performance trade-offs.

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) — Nov 2025

- Designed and executed machine learning experiments to evaluate Vision Transformer performance across sequential learning tasks.
- Developed automated Python-based training and evaluation pipelines for comparative model analysis.
- Analyzed model performance using quantitative evaluation metrics to measure knowledge retention and catastrophic forgetting.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems. .
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.
- **Technologies :** Python, PyTorch, Vision Transformers, Computer Vision

Machine Learning for Air Pollution [IEEE Publication] — May 2023

- Published an IEEE conference paper reviewing machine learning approaches for air quality prediction.
- Analyzed six machine learning models including Random Forest, SVM, Genetic Algorithms, LS-SVM, Decision Trees, and Artificial Neural Networks.
- Evaluated prediction-based and knowledge-discovery methodologies across six NAAQS pollutants.
- Identified real-time monitoring and IoT integration as important future directions for environmental analytics.